



RCD10.5XS power consumption

This document is designed to remove confusion as to what power you can expect to consume running on 12 volts with the RCD10.5XS upright two door compressor fridge – freezer.

Rated power: 9.5 Amps

In the case of the **RCD10.5XS** this should be seen as the maximum power that the refrigerator will consume from the battery at any moment. The highest power consumption occurs at start up when the refrigerator is warm, but once down to temperature the refrigerator compressor will consume less power.

As this above figure is shown in Dometic official documentation – people may think that that is an average hourly consumption figure, but it is not. The average consumption figure is much lower.

What is important to know and understand:

The average hourly power consumption over a 24-hour period.

Understanding this allows one to size their battery and solar appropriately so that the refrigerator does not run out of power/flatten the batteries.

The average consumption can vary with the ambient temperature; at an ambient of 30 Celsius, the refrigerator needs to consume more power than when the ambient is 25 Celsius. There are also other variables such as wiring size impacting on voltage loss, humidity, stock in the refrigerator, how often and how long the door is opened, how cold it is set and so on.

Due to vacuum panel assisted insulation and other considerations such as size of the condenser in this model, these are very efficient refrigerators and the cycle on times are relatively short, so when the compressor is cycled off as the refrigerator is cold enough, there is virtually no power consumption at all – just milliamps to power the electronics. (Let's call it "standby" consumption – same as your TV at home when off.)

It is not feasible for a manufacturer to provide precise consumption figures for all variable situations, but below information will help determine what is needed.

From Dometic production plant testing on RCD10.5XS:

With an ambient temperature of **27C**, with the temperature setting on setting **3**, with a full battery at **13.29 volts**, the refrigerator will consume **1.8 amperes per hour** on average whilst those conditions remain. That means in the unlikely event that the ambient temperature did not change from 27C for 24 hours – the refrigerator would consume $24 \times 1.8\text{amps} = 43.2$ amperes from the battery for that day.

If the Ambient was 32C for the daylight 12 hours and 22C for the night 12 hours, that means the average temperature for that day was 27C so – dependent still on other variables - the refrigerator should have consumed around 45 amperes from the battery storage for that day.

Tip; As all refrigerators need to run more in hot climates – it pays to allow a little extra. Ensure that you have enough for pretty much all Australian areas by assuming 2.3 – 2.5 amps per hour, size your batteries and solar array accordingly, and you are unlikely to run out of power due to the refrigerator consuming too much.